

Wildfire Ignition Risk Prediction (Colorado)

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Abstract

Wildfires are becoming a serious concern in Colorado because they affect communities, infrastructure, and the environment. Predicting the exact time and location of a wildfire ignition is very difficult, so this project focuses on ranking locations by wildfire ignition risk instead of predicting exact fire events.

This project combines wildfire ignition data from NASA’s MODIS FIRMS system with daily climate data from the gridMET dataset. The final dataset includes grid-level observations across Colorado from 2018 to 2023. Exploratory data analysis showed that wildfire events are rare, seasonal, and more closely related to accumulated dryness than single-day weather conditions.

Several machine learning models were tested, including Logistic Regression, Random Forest, Gradient Boosting, and XGBoost. The models were trained on data from 2018 to 2022 and evaluated on unseen data from 2023. Since wildfire events are rare, the focus was placed on risk ranking rather than traditional accuracy.

The results suggest that climate-based features can help identify areas with higher wildfire ignition risk. This approach can support wildfire monitoring, planning, and decision-making.

I. INTRODUCTION/BACKGROUND

Wildfires are a recurring problem in the western United States, including Colorado. Climate conditions such as high temperature, low precipitation, and seasonal dryness can increase the chance of wildfire ignition. Because of this, it is important to identify areas where wildfire risk may be higher.

Traditional wildfire risk methods often depend on past fire locations or static risk maps. However, fire risk changes over time because weather and climate conditions change. This project uses a data-driven approach to study wildfire risk using climate data and satellite fire detection data.

This project also connects to several important data science skills, including data preparation, feature engineering, exploratory analysis, machine learning, model evaluation, and reporting.

II. PROBLEM STATEMENT

The main problem addressed in this project is how to identify and rank locations in Colorado based on wildfire ignition risk using climate data. Since wildfire ignitions are rare and affected by many factors, predicting the exact location and timing of each fire is not realistic.

Instead, this project focuses on relative risk ranking. The goal is to find which locations are more likely to experience wildfire ignition compared to others. This is useful because even if exact prediction is difficult, risk ranking can still help with monitoring and resource planning.

III. RELATED WORK

Previous wildfire research has shown that climate conditions, drought, fuel dryness, and seasonal patterns are important factors in wildfire activity. Many studies use climate variables and historical fire records to model fire occurrence or burned area.

However, wildfire prediction is difficult because fire events are rare and unevenly distributed. Traditional accuracy can be misleading in this type of problem because a model could predict “no fire” most of the time and still appear accurate. Because of this, this project focuses more on ranking high-risk areas rather than only measuring accuracy.

IV. METHODOLOGY/APPROACH

This project follows a supervised machine learning approach to model wildfire ignition risk using historical climate and fire data. Wildfire ignition data was obtained from NASA’s MODIS FIRMS dataset, which provides satellite-based fire detections as geographic point locations. Climate data was obtained from the gridMET dataset, which provides daily gridded temperature and precipitation variables across the United States.

Fire detections were first filtered to Colorado and then spatially aligned to the gridMET climate grid by mapping each fire detection point to its corresponding grid cell. The data was aggregated at the grid-cell daily level, and a binary target variable was created where fire = 1 indicates at least one fire detection in that grid cell on that day, and fire = 0 otherwise.

Climate features included daily precipitation, maximum and minimum temperature, lagged precipitation values (1, 3, and 7 days), rolling precipitation sums (7 and 14 days), and a day-of-year variable to capture seasonality. Lagged and rolling features were constructed using only past information to prevent temporal data leakage.

Several machine learning models were evaluated, including Logistic Regression, Random Forest, Gradient Boosting, and XGBoost. Logistic Regression was used as a baseline model

with class-weight balancing to address extreme class imbalance. Tree-based models were selected to capture nonlinear relationships and interactions between climate variables identified during exploratory analysis.

Models were trained on data from 2018 to 2022 and evaluated on unseen data from 2023 to assess temporal generalization. Model performance was evaluated using ranking-based metrics rather than traditional accuracy.

V. DATA ANALYSIS

Exploratory data analysis revealed that wildfire ignitions are extremely rare in the dataset, occurring in approximately 0.01

EDA also showed strong seasonality in wildfire ignitions, with higher activity during warmer and drier periods. Accumulated dryness, represented by lagged and rolling precipitation features, showed a stronger relationship with wildfire ignition than single-day precipitation or temperature alone. These findings guided both feature selection and model choice.

For model evaluation, Logistic Regression, Random Forest, Gradient Boosting, and XGBoost were trained on historical data from 2018 to 2022 and evaluated on unseen 2023 data. Performance was primarily assessed using ROC-AUC to measure overall risk-ranking quality, and PR-AUC to evaluate performance on the rare fire class.

Tree-based models outperformed Logistic Regression, indicating that wildfire risk depends on nonlinear interactions between climate variables such as seasonality and accumulated dryness. Gradient Boosting achieved the strongest risk-ranking performance on unseen data.

To assess practical usefulness, risk scores were used to rank grid-cell locations by wildfire ignition risk. Results showed that a small fraction of high-risk locations captured a disproportionately large share of wildfire ignitions, demonstrating the value of risk ranking for prioritization.

VI. EXPECTED OUTCOMES

The expected outcome of this project is a practical method for ranking wildfire ignition risk across Colorado. The goal is not to predict every fire exactly, but to identify higher-risk locations more effectively than random selection.

The results are expected to show that climate variables can provide useful information for wildfire risk ranking. This type of approach could support wildfire monitoring, resource allocation, and early planning.

VII. CONCLUSION

This project demonstrates that wildfire ignition risk can be effectively studied using climate data and machine learning techniques. Because wildfire ignitions are rare and difficult to

predict exactly, framing the problem as a risk-ranking task provides a more practical and reliable approach than binary prediction.

By combining satellite-based fire detections with climate features such as accumulated dryness and seasonality, the models were able to generalize to unseen future data and identify high-risk locations more effectively than random selection. Tree-based models performed particularly well due to their ability to capture nonlinear relationships between climate variables.

Overall, this work highlights how data science methods can be applied to environmental risk problems and supports the use of climate-based risk ranking as a decision-support tool for wildfire monitoring and planning.

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